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Editorial Office
Solar Energy
Elsevier

Dear Editor,

We submit our manuscript **"Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin"** for consideration as a Research Article in *Solar Energy*. It is not under consideration elsewhere. The paper is about what a semi-transparent photovoltaic module should absorb, what it should hand to the crop below, and what that allocation is worth in water, carbon, and money — solar energy conversion and spectral management at the heart of the journal's scope.

Central question and significance. Can spectral co-design guided by non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (power + water + carbon + diagnostics) in a field-constrained agrivoltaic stack, and which barriers remain physical rather than economic? Static semi-transparent modules resolve the food–energy conflict only by uniformly sacrificing output or yield. Here an organic photovoltaic layer coupled to a nanoparticle-on-mirror (NPoM) cavity shapes the transmitted spectrum actively: the quantum channel buys its diagnostics out of the near-infrared that even plant-transparent designs divert or discard, and the trade-off is quantified rather than asserted.

What the analysis shows (all modeled, fully traceable).

- The design law: a 91.8 % local transport penalty, confined to a 1 % sentinel area, costs 0.9 % of canopy-scale device yield ($0.980 \to 0.971$) and adds single-complex SERS diagnostics at no extra land.
- The dual-band passbands at 750 nm and 820 nm lie outside PAR, so the photobiology and the crop-light budget do not compete; the land-use ledger returns a land-equivalent ratio of ≈ 1.4.
- The shield cuts crop evapotranspiration by 28 % ($460\,\mathrm{L\,m^{-2}\,yr^{-1}}$) and yields a Net Ecological Benefit of $72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$; for a $500\,\mathrm{m^2}$ cooperative module in Cameroon, payback is 4.32 yr at 30 % subsidy (2.68 yr with local materials).
- Durability is the leading coefficient, not a footnote: reported outdoor organic-PV lifetimes bound a five-year replacement cycle, re-run through the carbon and payback accounts.

Why *Solar Energy*? The quantum methods appear only as the design engine for the module's spectral architecture; every claim that reaches a decision is optical, agronomic, or economic, and no revenue line in the techno-economics depends on the quantum gain. The paper is deliberately falsifiable: its exposure to the live debate over electronic coherence is quantified (all carried numbers survive the recent reattribution of long-lived signals to vibronic channels), and the Discussion names the three experiments that would change its conclusions.

Novelty and positioning. To our knowledge, this is the first agrivoltaic stack whose spectral architecture is co-designed with numerically exact open-quantum-system dynamics, the first quantification of the local-to-global transport trade-off that makes a plasmonic diagnostic layer viable at canopy scale, and the first multifunctional TEA/LCA internalizing power, water, carbon, and sensing in a single account. Economic barriers are shown to be tractable; physical barriers (sub-2 nm cavity gaps, sentinel fouling, encapsulation lifetime) are stated honestly as the binding constraints.

Compliance. The manuscript has been language-edited; all authors approve the submission and declare no competing interests. An earlier version was considered at *Nature Energy*; we submit here as the more suitable venue. Subscription route (no APC). Corresponding author based in Cameroon.

Suggested reviewers: A. Olaya-Castro (UCL), J. Yuen-Zhou (UCSD), J. J. Baumberg (Cambridge), G. A. Barron-Gafford (Arizona)—covering quantum biology, polaritonic chemistry, NPoM/SERS, and agrivoltaics.

Sincerely,

Steve Cabrel Teguia Kouam (on behalf of all authors)

Enclosures: Manuscript, Supplementary Material, Highlights, Graphical Abstract.